AN ODE TO EUCLID

Then, my noble friend, geometry will draw the soul towards truth, and create the spirit of philosophy, and raise up that which is now unhappily allowed to fall down.

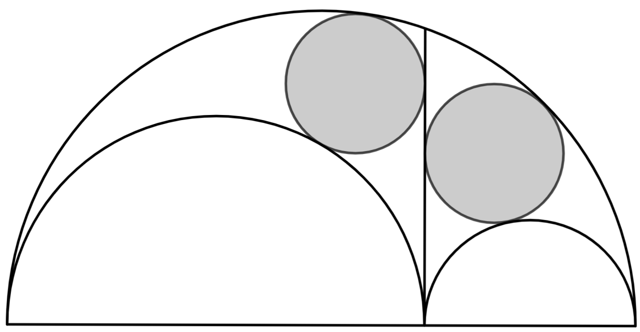
Nothing will be more likely to have such an effect.

Then nothing should be more sternly laid down than that the inhabitants of your fair city should by all means learn geometry.
Plato. The Republic VII.

And since goodness is distinct from beauty (for it is always in actions that goodness is present, whereas beauty is also in immovable things), they are in error who assert that the mathematical sciences tell us nothing about beauty or goodness; for they describe and manifest these qualities in the highest degree, since it does not follow, because they manifest the effects and principles of beauty and goodness without naming them, that they do not treat of these qualities.
Aristotle. Metaphysics XIII.

They Came, he used to say, every one of them, in the conviction that they would get from the lectures some one or other of the things that the world calls good; riches or health, or strength, in fine, some extraordinary gift of fortune. But when they found that Plato's reasonings were of sciences and numbers, and geometry, and astronomy, and of good and unity as predicates of the finite, methinks their disenchantment was complete.
Aristoxenus. Elements of Harmony II.

Why was Plato so keen for people to learn geometry? Was is it for reasons of utility? To sharpen young people's facilities for abstract thought? These notes are not an attempt to answer this question. They are an attempt to showcase a little bit of the geometry itself.



Archimedes's Twin Circles

It is presumed that Euclid of Alexandria c. 300 BC had studied with students of Plato, if not at Plato's Academy itself [2]. He is famous for compiling the most successful textbook ever written, *The Elements*, in which he delicately builds the geometry that bears his name from a handful of assumed and unproven statements – his Postulates.

1. Euclid's Postulates

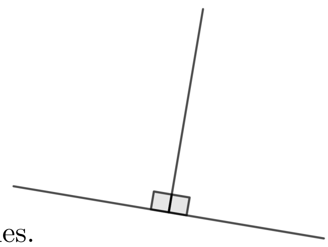
Postulate 1: It is possible to draw exactly one line through any two points.

By *line* we mean a straight line that extends indefinitely in both directions. We are usually only concerned with finite portions of lines that lie between two end points. We call these *line segments* and use the notation AB to refer to both the *length* of the segment between points A and B as well as the segment itself. Now there is no notion of absolute length in Euclid's geometry but we can make intuitive sense of 'longer than', or 'shorter than', as meaning one line segment includes, or is included by, another. What's more, given a fixed segment PQ, we may declare this to be a *unit* segment and then understand the length of any other segment as the number of copies of PQ that fit into it. In this way numbers can be associated with lengths – a topic we shall return to later.

Postulate 2: It is possible to extend any given line segment to a segment of any length.

The next postulate concerns circles – those points that lie a given fixed distance (the radius) from a given fixed point (the centre).

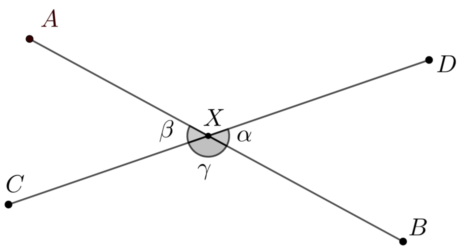
Postulate 3: Given two points, it is possible to draw a circle with centre at the first that passes through the second.



Angles occur where lines intersect. We denote the angle at the intersection X of segments AX and BX by $\angle AXB$, or simply $\angle X$ or a Greek letter if it is unlikely to cause confusion. As with lengths, we can compare the relative sizes of angles and make sense of 'bigger than/smaller than' as meaning includes/is included by. However, whereas we have no notion of absolute length, the next postulate gives us a canonical measure for angles.

If a line is extended from a point on a line such that the angles made on either side are equal, we call these *right angles*.

Postulate 4: All right angles are equal to one another.



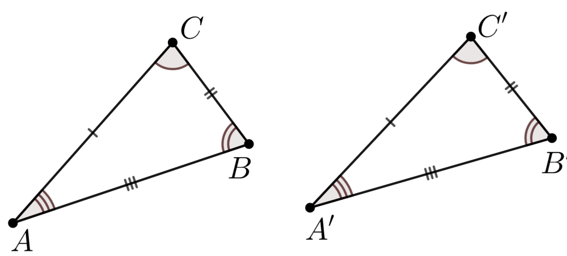
An important consequence of Postulate 4 is that if pairs of angles are related as α is to β (sometimes called 'opposite angles') at an intersection of two lines, then they are equal.

Proof. In the diagram $\beta + \gamma$ is equal to two right angles since AXB is a straight line. Likewise $\alpha + \gamma$ is also equal to two right angles, since CXD is a straight line. By Postulate 4 these two sums are equal and therefore $\alpha = \beta$. $\square$

Our next postulate is not technically one of Euclid's but it is of crucial importance for reasoning about triangles. It gives us a way of determining that two given triangles are essentially the same.

We call two triangles ABC and $A'B'C'$ *congruent* if their sides and angles are pair-wise equal.

$$AB = A'B', \ \ BC = B'C', \ \ AC = A'C'$$
$$\angle A = \angle A', \ \ \angle B = \angle B', \ \ \angle C = \angle C'$$



SAS Postulate (Side-Angle-Side): A given triangle ABC is congruent to a second triangle if and only if there is a way of labelling the vertices of the second triangle $A'B'C'$ so that

$$AB = A'B', \quad AC = A'C', \quad \angle BAC = \angle B'A'C'.$$

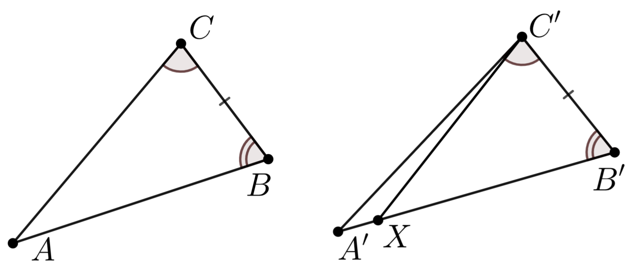
Euclid presents the SAS Postulate as his fourth *Proposition* – a logically derived consequence of his postulates. Modern standards of rigour however deem his organisation of the subject to be somewhat flawed and so to avoid getting bogged down with knotty technicalities, we instead state the result as a Postulate. We are after all in the business of doing geometry, not logical analysis.

Our first Proposition gives another very useful way of deducing that two triangles are congruent.

Proposition 1. ***ASA (Angle-Side-Angle)***: *A given triangle ABC is congruent to a second triangle if there is a way of labelling the vertices of the second triangle $A'B'C'$ so that*

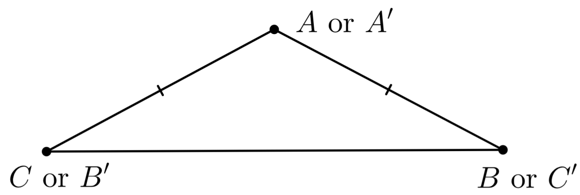
$$BC = B'C', \quad \angle ABC = \angle A'B'C', \quad \angle BCA = \angle B'C'A'$$

Proof. Mark point X on line $A'B'$ such that $BA = B'X$. Then it follows from **SAS** that ABC is congruent to $XB'C'$. So $\angle B'C'X = \angle BCA = \angle B'C'A'$ and X must lie on $A'C'$. Since X also lies on $A'B'$ it must equal A'. Hence ABC is congruent to $A'B'C'$. □



An important consequence of **SAS** and **ASA** is the familiar fact that given a triangle ABC, we have that $AB = AC$ if and only if the angle at B is equal to the angle at C.

Proof. The idea, which is somewhat abstract so may require a little thought, is to apply **SAS** and **ASA** to the triangle $A'B'C'$ given by $A' = A$, $B' = C$ and $C' = B$.



If we assume that $AB = AC$, then $A'C' = AB = AC = A'B'$ and we know $\angle BAC = \angle B'A'C$ so we may deduce from **SAS** that ABC and $A'B'C'$ are congruent. And in particular that the angle at B is equal to the angle at C.

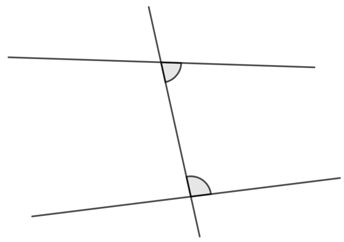
If on the other hand we assume that $\angle BAC = \angle B'A'C$, then $\angle A'C'B' = \angle ACB = \angle ABC = \angle A'B'C'$ and we know that $BC = B'C'$ so we may deduce from **ASA** that ABC and $A'B'C'$ are congruent. And in particular that $AB = AC$. □

Using just what he have built so far, we are more-or-less[1] in a position to demonstrate the validity of some fundamental 'ruler and compass constructions', no doubt remembered from school:

- the bisection of an angle
- drawing a line perpendicular to a line through a given point on the line
- drawing a line perpendicular to a line through a given point not on the line
- identifying the midpoint of a given segment

The 5^{th} and final of Euclid's postulates is peculiarly complicated in comparison to the previous four. It has rich historical significance we won't comment on further other than to say that for centuries a great deal of effort went into trying to prove that it is a necessary consequence of the others. Mathematicians gradually came to the conclusion that there are perfectly consistent non-Euclidean geometries that are worthy of study for which it does not hold. Thus the 5^{th} Postulate is what gives Euclidean geometry its distinct character. Its story is a good case study in motivating the need for rigorous proofs and why we shouldn't always rely on our intuitions.

Postulate 5 (Parallel axiom): If a straight line cuts two straight lines so that the interior angles on one side add up to less than two right angles, then those two lines meet on that side.

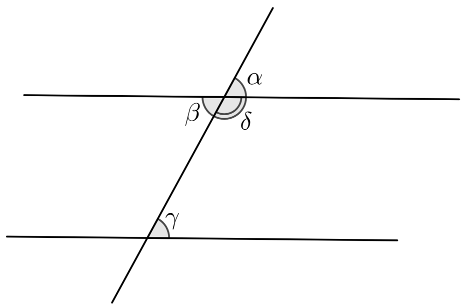


This final postulate implicitly assumes that our geometry is that of a flat 2-dimensional plane. In 3 dimensions there is extra freedom for lines to not meet and yet not be parallel but since we are concerned with the geometry of the plane, for us, lines are by definition *parallel* if they do not meet.

Let us state a few consequences that we think deserve the title of Proposition.

Proposition 2. *If a pair of parallel lines are crossed by a third line as shown, then the angles marked α, β, γ are all equal.*

Proof. We have already seen that it follows from Postulate 4 that $\alpha = \beta$ so we just need to show that $\alpha = \gamma$ also. It follows from the Postulate 5 that $\gamma + \delta$ is equal to two right angles for if $\gamma + \delta$ was less, the lines would meet on this side and if it were greater, they would meet on the other (why?). But we know that $\alpha + \delta$ is also equal to two right angles because they form a straight angle. Therefore $\alpha + \delta = \gamma + \delta$ and hence $\alpha = \gamma$, as required. □

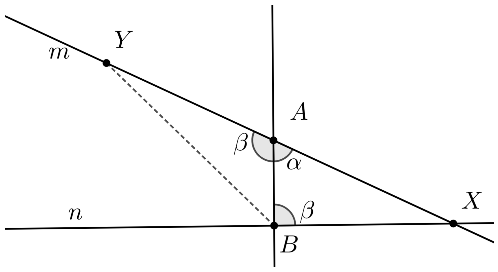


Proposition 3. *Distinct lines m, n are parallel if and only if there is some line that meets both such that interior angles α and β on one side add to exactly two right angles.*

Proof. We first show the easier implication that if m, n are parallel then some 'transverse' line is such that the interior angles on one side add to two right angles. In fact for any transverse line, we showed in the proof of Proposition 2 that the internal angles must sum to two right angles, else m, n would meet and not be parallel.

[1] *Hint for the readers who wishes to fill in the remaining gap:* The key is to show that if ABC and $A'BC$ are both isosceles triangles with A and A' on the same side of BC, then the line joining AA'

- bisects the angles at A and A'
- is perpendicular to BC
- meets the line BC at the midpoint of BC

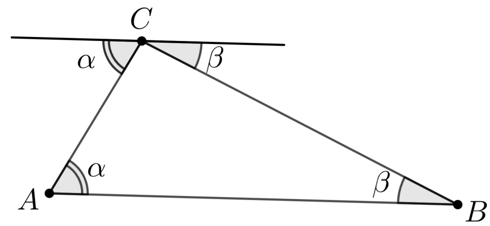


For the reverse implication, suppose we have a pair of lines m, n and some transverse line as depicted such that angles $\alpha + \beta$ equals two right angles. If m, n were *not* parallel then they meet at some point X, say. Let Y be the point on m on the opposite side of the transverse line to X such that $AY = BX$. Then triangles ABX, BAY are congruent by **SAS** since $AB = BA, AX = BY$ and $\angle BAX = \angle ABY = \beta$. But then $\angle BAX = \angle ABY = \alpha$. So Y lies on m and therefore both m, n pass through X and Y. Postulate 1 implies this conflicts with the assumption that m, n are distinct lines and we must conclude that m, n are in fact parallel. □

The significance of Proposition 3 is that it allows us to add to our previous list of valid constructions that of drawing a line (*the* line) parallel to another through a given point. For example by drawing a perpendicular from the point to the line and then a perpendicular from the point to the new line.

Proposition 4. *The angles of a triangle sum to a straight angle.*

Proof. Construct the line parallel to AB that passes through C. It follows from Proposition 1 that the angles marked with the same letters are equal which proves the result because the three angles internal angles together form a straight angle. □

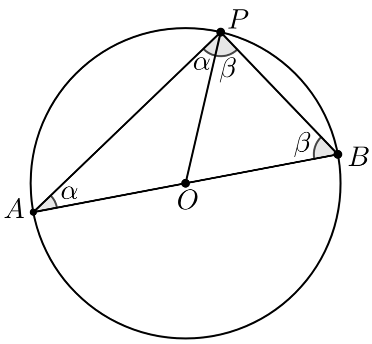


2. Thales and Pythagoras

It is often said that philosophy began with Thales of Miletus and that he was the first true mathematician – the 'originator of the deductive organization of geometry' [2].

Theorem 1 (Thales's Theorem)**.** *Let AB be the diameter of a circle and P some point other than A, B on the circle. Then the angle $\angle APB$ is a right angle.*

Proof. Draw the line through P and the centre of the circle O. From the definition of a circle, this creates two triangles, each with a pair of equal sides. We have seen that this implies the opposite angles are also equal. Since the sum of the angles of triangle APB is $2\alpha + 2\beta$, and this is equal to two right angles by Proposition 4, we conclude that $\angle APB = \alpha + \beta$ is equal to one right angle, as required. □



Before turning to Pythagoras's Theorem we need the notion of *area* and prove the following basic and most important result about the area of a triangle. We assume an intuitive notion of area that respects the basic principle, akin to lengths and angles, that if one shape in entirely included in another, it's area is at most that of the shape that includes it. We shall take for granted that congruent shapes have equal area, that shapes composed of multiple non-overlapping parts have area equal to the sum those parts, and that the area of a rectangle[2] is the length of one side times an adjacent side – it's 'height' times 'width'.

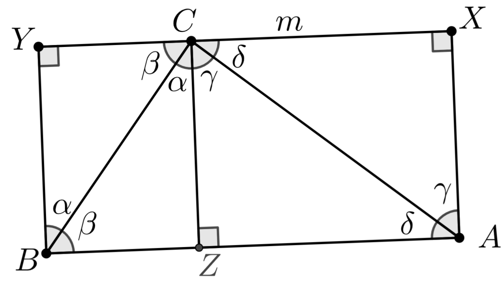
[2]A *rectangle* is a four sided shape in which all four angles are right angles. In the course of proving our next proposition, it will be shown that pairs of opposite sides in a rectangle are equal in length.

Proposition 5. *Twice the area of triangle ABC is equal to the side length AB times the length of the perpendicular from C to the line through AB.*

Note that there is a unique line perpendicular to AB through C so the perpendicular distance is also uniquely defined.

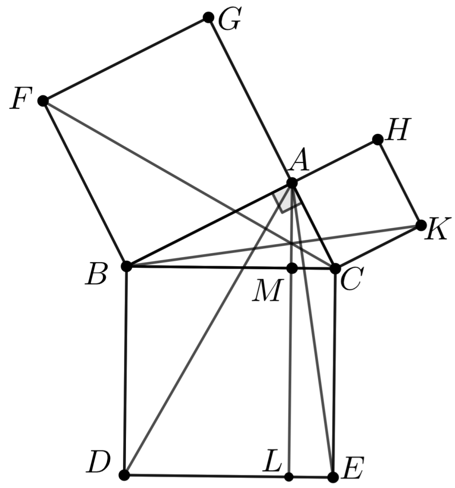
Proof. Let the perpendicular through C meet AB at Z. We shall treat the case where Z lies between A and B and leave the case when it is outside for the reader to consider.

Label by m the line parallel to AB through C. Mark the points X, Y on m such that AX, BY are perpendicular to AB so that we have 3 rectangles drawn. By Propositions 2 and 4, the angles marked with the same letter are equal. The result would follow if we could show that triangles AZC is congruent to CXA, and that triangle ZBC is congruent to YCB. But these congruences follow from the **ASA** condition, so the proof is complete.



We are now finally ready to present Euclid's proof of Proposition 47 from his Elements Book 1.

Theorem 2 (Pythagoras's Theorem)**.** *If triangle ABC has a right angle at A then $AB^2+AC^2 = BC^2$.*



Proof. Construct the squares on the sides of ABC as shown. Draw the line through A perpendicular to BC and mark the points M, L as shown. First note that triangles ABD and FBC are congruent by **SAS** because $AB = FB$, $BC = DB$ and $\angle ABD = \angle FBC = \angle ABC +$ right angle. So the area of the square on AB has area equal to rectangle $BDLM$, since the former is twice the area of triangle FBC by Proposition 5 (the 'perpendicular height' being AB) and the latter is twice that of ABD. A symmetric argument implies that the square on AC has area equal to rectangle $CELM$. Together these rectangles form the square on BC and the proof is complete. □

With his proof of Pythagoras's Theorem, Euclid has brought us to the summit of Book I. From here we are able to see down to an unsettling realisation that brought shame to Plato's 'Athenian' in Laws VII: *"I was ashamed, not only of myself, but of all the Greek world."*

3. From Pythagoras (fl. 530 BC) to Eudoxus (fl. 370 BC)

The three great problems of early Greek geometry all concern ruler and compass constructions.

- To construct a square with area equal to that of a given circle.
- To trisect a given angle.
- Given the side of a cube, to construct the side length of one with double the volume.

It is curious how fruitful these problems have been given that we now know that they are impossible – that no ruler and compass constructions exist for any of these problems. They remind us that progress sometimes consists of *explanations of failure.* In fact, progress of this type is hardly exceptional. The German mathematician David Hilbert gave another nice example of this phenomenon in his famous speech titled Mathematical Problems.

"After seeking in vain for the construction of a perpetual motion machine, the relations were investigated which must subsist between the forces of nature if such a machine is to be impossible; and this inverted question led to the discovery of the law of the conservation of energy." (Paris, 1900)

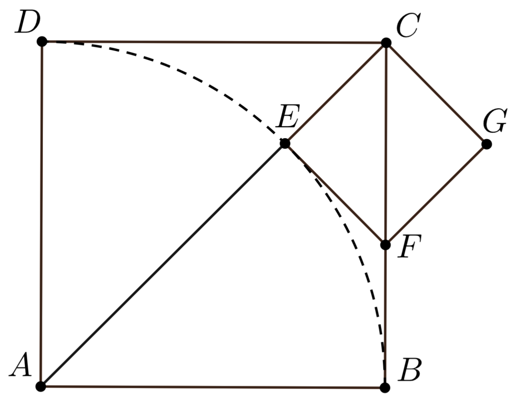
A lot of very good maths was inspired by these three problems and, eventually, by proofs and explanations for why they are impossible. Such maths is beyond the scope of what we are able to cover here but there is another famous 'failure' we can explain. Plato writes in Laws VII:

Athenian One ought to declare, then, that the freeborn children should learn as much of these subjects as the innumerable crowd of children in Egypt learn along with their letters. First, as regards counting, lessons have been invented for the merest infants to learn, by way of play and fun,—modes of dividing up apples and chaplets, so that the same totals are adjusted to larger and smaller groups, and modes of sorting out boxers and wrestlers, in byes and pairs, taking them alternately or consecutively, in their natural order. Moreover, by way of play, the teachers mix together bowls made of gold, bronze, silver and the like, and others distribute them, as I said, by groups of a single kind, adapting the rules of elementary arithmetic to play; and thus they are of service to the pupils for their future tasks of drilling, leading and marching armies, or of household management, and they render them both more helpful in every way to themselves and more alert. The next step of the teachers is to clear away, by lessons in weights and measures, a certain kind of ignorance, both absurd and disgraceful, which is naturally inherent in all men touching lines, surfaces and solids.
Clinias What ignorance do you mean, and of what kind is it?
Athenian My dear Clinias, when I was told quite lately of our condition in regard to this matter, I was utterly astounded myself: it seemed to me to be the condition of guzzling swine rather than of human beings, and I was ashamed, not only of myself, but of all the Greek world.
Clinias Why? Tell us what you mean, Stranger.
Athenian I am doing so. But I can explain it better by putting a question. Answer me briefly: you know what a line is?
Clinias Yes.
Athenian And surface?
Clinias Certainly.
Athenian And do you know that these are two things, and that the third thing, next to these, is the solid?
Clinias I do.
Athenian Do you not, then, believe that all these are commensurable one with another?
Clinias Yes.
Athenian And you believe, I suppose, that line is really commensurable with line, surface with surface, and solid with solid?
Clinias Absolutely.
Athenian But supposing that some of them are neither absolutely nor moderately commensurable, some being commensurable and some not, whereas you regard them all as commensurable,—what do you think of your mental state with respect to them?
Clinias Evidently it is a sorry state.
Athenian Again, as regards the relation of line and surface to solid, or of surface and line to each other—do not all we Greeks imagine that these are somehow commensurable with one another?
Clinias Most certainly.
Athenian But if they cannot be thus measured by any way or means, while, as I said, all we Greeks imagine that they can, are we not right in being ashamed for them all, and saying to them, "O most noble Greeks, this is one of those 'necessary' things which we said it is disgraceful not to know, although there is nothing very grand in knowing such things."

What is meant here by commensurable and why the sorry mental state? As mentioned earlier, there is no notion of absolute size to measure lengths in Euclid's geometry. Numbers enter as a means of comparing one line segment to another (or the area/volume of one shape to another). The length of a diameter of a circle is *two* times its radius for example because two copies of the radius exactly cover a diameter. In general, we said that we can make sense of the length of the segment AB by asking how many times a given *unit* segment PQ fits into it. If the answer is not a whole number of times, we may need to resize our measuring segment PQ. Thus the Pythagoreans sought to understand how the side length of a square compares to its diagonal by constructing a (possibly very small) measuring segment PQ that can be used to simultaneously measure these two lengths. They failed in their attempt to do so because there is no such segment!

Theorem 3. *Let $ABCD$ be a square. The ratio $AC : AB$ cannot equal a ratio of whole numbers.*

Proof. Suppose instead that there *is* some small line segment PQ such that the side length AB and diagonal AC are both measurable in terms of PQ. That is, there are some positive integers m and n such that $AB = m \times PQ$ and $AC = n \times PQ$. We are going to show that this is impossible.



Let E be the point on AC such that $AE = AB$. Construct the segment EF perpendicular to AC that meets BC at F. Now ABC has two sides equal, therefore has two angles equal to half a right angle. So CEF also has two angles equal to half a right angle and therefore $EF = CE$. Complete the square $CEFG$ with the point G. Now it turns out we can measure the side and diagonal of this new square in terms of PQ too. First note that Pythagoras' Theorem applied to the two right triangles AFE and AFB implies that $EF = BF$. So we have

$$EC = AC - AE = AC - AB = (n - m) \times PQ$$

and

$$CF = BC - BF = BC - EF = BC - EC = m \times PQ - (n - m) \times PQ = (2m - n) \times PQ.$$

Let us take stock of what we have done. We started with a square $ABCD$ and supposed the existence of some segment PQ that measured both the side and diagonal of this square. We then constructed a smaller square whose side and diagonal can also be measured with PQ. But this new square is less than half the size of the original. If we did the same thing again, starting with $CEFG$ in place of $ABCD$, we'd have a square that was less than one quarter the size of the original with side and diagonal still both measurable by PQ. Continuing in this we would eventually have a square with side length less than PQ and yet was still somehow measurable in terms of PQ. This is clearly impossible, and so no such PQ can exist. $\square$

So the diagonal of a square is incommensurate with its side. This realisation shook the Pythagoreans who had taken for granted that two magnitudes of like kind (two lengths, two areas or two volumes) could be compared in this way. In the absence of a common way of measuring the lengths, what does the ratio of the diagonal to the side length even *mean*?

The way forward was shown by Eudoxus of Cnidus, a contemporary of Plato, around 370 BC. He realised that what really mattered about ratios is not what they *are* or what they *mean*, but *how they* *behave.* In this regard, it is interesting to note how Eudoxus anticipates the philosophical ideas later captured by the aphorism: *"meaning as use"*. Here is how he characterised the concept of ratio.

Let a, b be two magnitudes of the same kind and c, d two other magnitudes of the same kind (possibly different to that of a, b). The ratio $a : b$ between magnitudes is equal to the ratio $c : d$ if for all pairs of *integers* m, n, the value $na - mb$ is positive, zero or negative exactly when $nc - md$ is correspondingly positive, zero or negative. Furthermore $a : b$ is not equal to $c : d$ if there is some pair of *integers* m, n for which $na - mb$ is positive while $nc - md$ is negative.

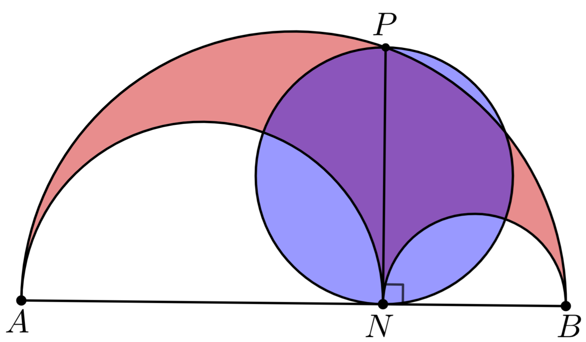
Thus Eudoxus *defined* the ratio of two quantities a, b of the same kind to be the pair $a : b$ together with this rule for talking about which of two ratios is bigger than another. It was an ingenious resolution to the problem of incommensurability because it not only provided a definition of ratio that salvaged the theory, but it is also managed to preserve the privileged status of whole numbers that was so important to the Pythagoreans.

4. Ephebic Training

At this point the reader may be inclined to think that maths is not much of a spectator sport. We agree. Even geometry. He or she is therefore invited to attempt the following exercises. This is their opportunity to get a *feel* for what Euclid was up to. We do encourage that the steps described in the first one at least be carried out (with actual ruler and compass!) for which we have not provided a diagram.

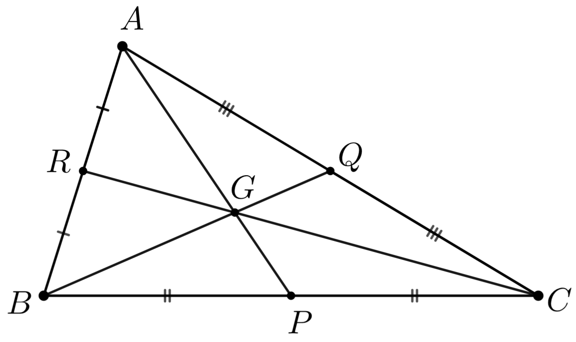
(1) Given a segment BC and a point A not on the line through BC, construct a point X such that $BC = AX$. *Solution:*
   (a) Draw the circle centre A, radius AB.
   (b) Draw the circle centre B, radius BA.
   (c) If we call one of the points of intersection of these two circles F, then the triangle FAB is equilateral.
   (d) Draw the circle centre B, radius BC. Extend the segment FB to meet this circle at G.
   (e) Draw the circle centre F, radius FG. Extend the segment FA to meet this circle at H.
   (f) Then $AH = BG = BC$, so we are finished.

(2) Consider a semicircle with diameter AB with a point N lying on AB and the two semicircles with diameters AN and NB lying inside the first. Show that the area enclosed by these three arcs (shown in red) is equal to the area of the circle with diameter NP (shown in blue) where NP is perpendicular to AB and P lies on the larger semicircle.



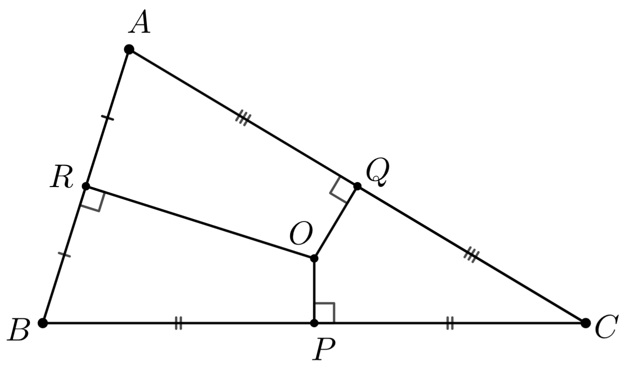
Hint: You may assume that the ratio of a circle's area to the square of its radius is the same for all circles.

(3) Let ABC be a triangle and let P, Q, R be the midpoints of the sides BC, CA, AB, respectively. It can be shown that the line segments AP, BQ and CR meet at a single point G, known as the *centroid* of ABC.



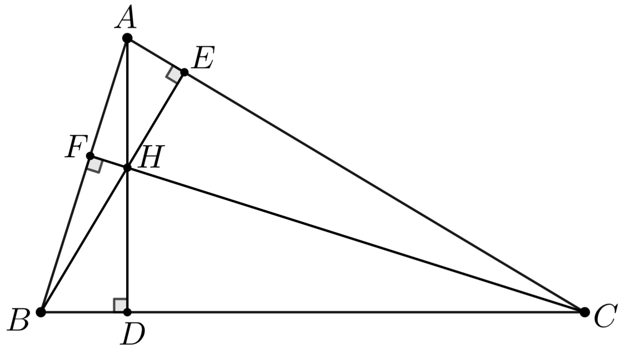
This exercise concerns a couple of other well known points associated to the triangle ABC, its *circumcenter* and *orthocenter*.

(a) Show that the three lines perpendicular to AB, BC, CA through R, P, Q respectively meet at a single point, O, the circumcenter.



Hint: This point O is equidistant to A, B, C. To get some purchase on the problem, define O to be the intersection of two of the three perpendiculars and then show that the line through O perpendicular to the third edge of ABC meets it at the desired point. Pythagoras' Theorem may be helpful.

(b) Let the line through A perpendicular to BC meet BC at D, the line through B perpendicular to CA meet CA at E and the line through C perpendicular to AB meet AB at F. Show that the three lines (extended if necessary) through AD, BE, CF meet at a single point H, the orthocenter.

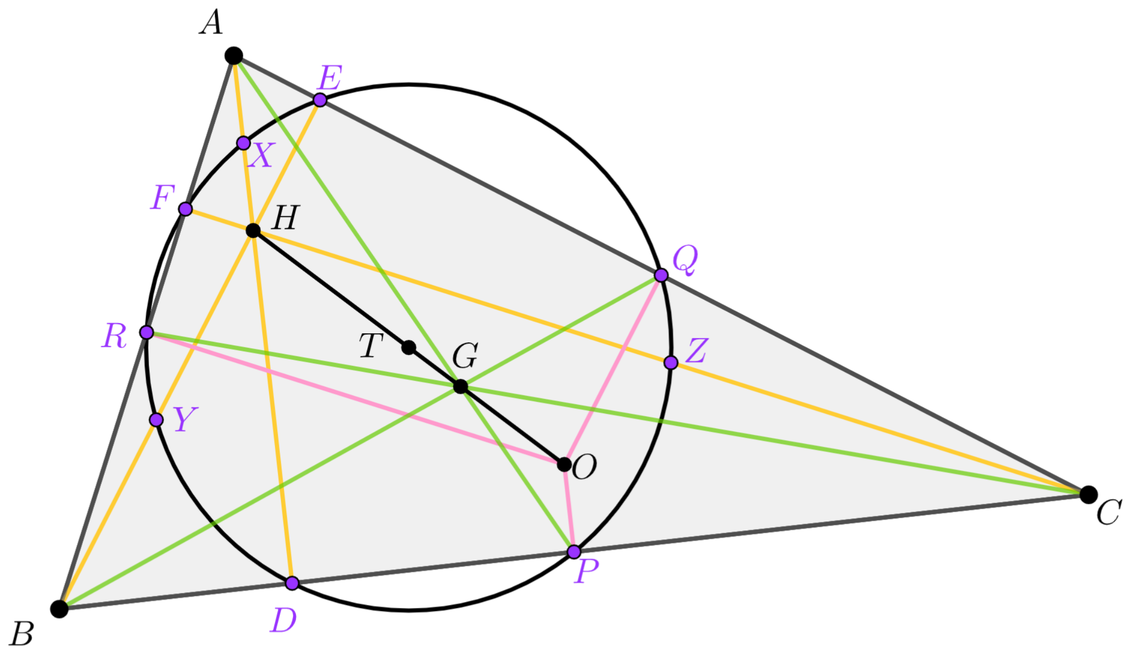


Hint: Consider the triangle formed by the lines through A, B, C parallel to BC, CA, AB, respectively.

* * *

We end with a treat. The three points O, G, H defined in exercise (3) are shown together in the figure below. The co-incidental lines that allow us to define these points are already curious but there is more of triangle ABC to be unearthed. Euler proved in 1765 the remarkable fact that O, G, H themselves always lie on a single straight line, known as the **Euler line**. He also showed that regardless of the initial relative positions of A, B, C, the point G always divides the segment OH in the precise ratio 1:2.

Recall also the points D, E, F and P, Q, R defined in exercise (3). We add to these 6 the midpoints X, Y, Z of AH, BH, CH, respectively, to get the 9 purple points in the figure. Quite miraculously these 9 points always lie on a single circle, aptly known as the **nine-point circle**. Even more, the centre T of this nine-point circle *also* lies on the Euler Line. Always lying at exactly the midpoint of OH.



The reader is once again invited and encouraged to get out his straightedge and compass. To carefully construct these nine points and watch how by his own hand O, G, H and T fall in line.

To see what Plato saw.

References

[1] A D Gardiner and C J Bradley, *Plane Euclidean Geometry: Theory and Problems*

[2] U C Merzbach and C B Boyer, *A History of Mathematics*

[3] M F Burnyeat, *Plato on Why Mathematics is Good for the Soul*